chapter 12 Statement of Cash Flows

Financial Accounting

12e

Libby • Libby • Hodge

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Learning Objectives

After studying this chapter, you should be able to:

12-1 Classify cash flow statement items as part of net cash flows from operating, investing, and financing activities.

12-2 Report and interpret cash flows from operating activities using the indirect method.

12-3 Analyze and interpret the quality of income ratio.

12-4 Report and interpret cash flows from investing activities.

12-5 Analyze and interpret the capital acquisitions ratio.

12-6 Report and interpret cash flows from financing activities.

12-7 Understand the format of the cash flow statement and additional cash flow disclosures.

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Understanding the Business

Positive cash flows permit a company to . . .

Pay dividends to owners.

Expand its operations.

Take advantage of investment opportunities.

Replace worn assets.

Both managers and analysts need to understand the various sources and uses of cash that are associated with business activity.

Some Wall Street analysts go so far as to say "**Cash flow is king**."

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Learning Objective 12-1

12-1 Classify cash flow statement items as part of net cash flows from operating, investing, and financing activities.

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What is cash?

The statement of cash flows explains how the amount of cash on the balance sheet at the beginning of the period has become the amount of cash reported at the end of the period.

The definition of cash includes cash and cash equivalents. Cash equivalents are short-term, highly liquid investments that are both:

- Readily convertible into known amounts of cash and
- So near to maturity there is little risk that their value will change if interest rates change.

Generally, only investments with original maturities of three months or less qualify as cash equivalents

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Exhibit 12.1

Consolidated Statement of Cash Flows

There are three categories of activities: operating, investing, and financing

The ending cash balance should agree with the balance sheet.

NATIONAL BEVERAGE CORP.
Consolidated Statement of Cash Flows*
Year Ended April 29, 2023
(dollars in thousands)

Cash flows from operating activities:	
Net income	$ 142,164
Adjustments to reconcile net income to net cash provided by (used in) operating activities:	
Depreciation and amortization	7,532
Changes in operating assets and liabilities:	
Accounts receivable	(11,326)
Inventories	9,740
Prepaid expenses	19,725
Accounts payable	(10,193)
Accrued expenses	9,195
Deferred income taxes—net	(4,009)
Net cash provided by operating activities	162,828
Cash flows from investing activities:	
Purchases of property, plant, and equipment	(21,979)
Proceeds from disposal of property, plant, and equipment	27
Purchase of short-term investments	(1,252)
Proceeds from sale of short-term investments	1,019
Net cash used by investing activities	(22,185)
Cash flows from financing activities:	
Repayment of principal on long-term debt	(30,000)
Proceeds from issuance of long-term debt	8,496
Repurchase of stock	–
Proceeds from issuance of stock	988
Payment of cash dividends	(10,103)
Net cash used by financing activities	(30,619)
Net Increase (Decrease) in Cash and Equivalents	110,024
Cash and Equivalents—Beginning of Year	48,050
Cash and Equivalents—End of Year	$158,074

*Certain amounts have been adjusted for pedagogical purposes.

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Classifications of the Statement of Cash Flows

Operating Activities → Cash inflows and outflows directly related to revenues and expenses reported on the income statement. There are two approaches for presenting this section of the statement.

Investing Activities → Cash inflows and outflows related to the purchase or disposal of long-lived productive asses and investments in the securities of other companies.

Financing Activities → Cash inflows and outflows related to exchanges of cash with creditors (debtholders) and owners (stockholders).

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Cash Flows from Operating Activities
Direct Method vs. Indirect Method (1 of 3)

There are two alternative approaches for presenting the operating activities section of the statement.

Direct Method

Reports the cash effects of each operating activity

Indirect Method

Starts with accrual net income and converts to net cash flow from operating activities

99% of large U.S. companies use the indirect method

The cash flows from operating activities are always the same, regardless of whether the direct or indirect method is used.

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Cash Flows from Operating Activities
Direct Method vs. Indirect Method (2 of 3)

The **direct method** reports the components of cash flows from operating activities as gross receipts and gross payments.

Inflows	Outflows
Cash received from:	**Cash paid for:**
Customers	Purchase of services (electricity, etc.) and goods for resale
Dividends and interest on investments	Salaries and wages
	Income taxes
	Interest on liabilities

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Cash Flows from Operating Activities
Direct Method vs. Indirect Method (3 of 3)

The two operating cash flow presentations are simply alternative ways to arrive at the same number.

Direct*		Indirect	
Cash collected from customers	$ 1,161,606	Net income	$ 142,164
Cash payments to suppliers	(776,596)	Depreciation and amortization	7,532
Cash payments for other expenses	(173,653)	Changes in operating assets and liabilities	13,132
Cash payments for interest	(242)		
Cash payments for income taxes	(48,287)		
Net cash provided by operating activities	$ 162,828	Net cash provided by operating activities	$ 162,828

*The direct amounts are estimated from National Beverage's disclosures.

Tip Cash flow from operating activities is the same regardless of whether it is computed using the direct or indirect method.

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Cash Flows from Investing Activities

Cash flows from investing activities are cash inflows and outflows related to the purchase and disposal of long-lived productive assets and investments in the securities of other companies.

Typical cash flows from investing activities include

Inflows	Outflows
Cash received from:	**Cash paid for:**
Sale or disposal of property, plant, and equipment	Purchase of property, plant, and equipment
Sale or maturity of investments in securities	Purchase of investments in securities

The difference between these cash inflows and outflows is called **net cash provided by (used by) investing activities.**

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Cash Flows from Financing Activities

Cash flows from financing activities include exchanges of cash with creditors (debtholders) and owners (stockholders).

Usual cash flows from financing activities include the following:

Inflows	Outflows
Cash received from:	Cash paid for:
Borrowing on notes, mortgages, bonds, and so on from creditors	Repayment of principal to creditors (excluding interest, which is an operating activity)
Issuing stock to owners	Repurchasing stock from owners
	Dividends to owners

The difference between these cash inflows and outflows is called **net cash provided by (used by) financing activities.**

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Net Increase (Decrease) in Cash

The combination of the net cash flows from operating, investing, and financing activities must equal the net increase or decrease in cash.

Net cash provided by operating activities	$162,828
Net cash used in investing activities	(22,185)
Net cash used in financing activities	(30,619)
Net increase in cash and equivalents	110,024
Cash—Beginning of Year	48,050
Cash—End of Year	$158,074

Beginning and ending balances from the balance sheet

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Relationships to the Balance Sheet and Income Statement (1 of 2)

- Preparing and interpreting the cash flow statement requires an analysis of the balance sheet and income statement accounts that relate to the three sections of the cash flow statement.
  - Companies cannot prepare the statement of cash flows using the amounts recorded in the T-accounts because those amounts are based on accrual accounting.
  - Companies must analyze the numbers recorded under the accrual method and adjust them to a cash basis.
- To prepare the statement of cash flows, they need the following data:
  - Comparative balance sheets
  - A complete income statement
  - Additional details concerning selected accounts

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Relationships to the Balance Sheet and Income Statement (2 of 2)

Preparing and interpreting the cash flow statement requires an analysis of the balance sheet and income statement accounts that relate to the three sections of the cash flow statement.

$$\triangle \text{Cash} = \triangle \text{Liabilities} + \triangle \text{Stockholders' Equity} - \triangle \text{Noncash Assets}$$

Derives from . . .

$$\text{Assets} = \text{Liabilities} + \text{Stockholders' Equity}$$

Thus, **any transaction that changes cash must be accompanied by a change in liabilities, stockholders' equity, or noncash assets.**

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Exhibit 12.2

Selected Cash Transactions and Their Effects on Other Balance Sheet Accounts

Category	Transaction	Cash Effect	Other Account Affected
Operating	Collect accounts receivable	+Cash	−Accounts Receivable (A)
	Pay accounts payable	−Cash	−Accounts Payable (L)
	Prepay rent	−Cash	+Prepaid Rent (A)
	Pay interest	−Cash	−Retained Earnings (SE)
	Sale for cash	+Cash	+Retained Earnings (SE)
Investing	Purchase equipment for cash	−Cash	+Equipment (A)
	Sell investment securities for cash	+Cash	−Investments (A)
Financing	Pay back debt to bank	−Cash	−Notes Payable–Bank (L)
	Issue stock for cash	+Cash	+Common Stock and Additional Paid-in Capital (SE)

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Preliminary Steps in Preparing the Cash Flow Statement

Any transaction that changes cash must be accompanied by a change in liabilities, stockholders' equity, or noncash assets.

Use the following preliminary steps to prepare the cash flow statement:

1. Determine the change in each balance sheet account.
2. Classify each change as relating to operating (O), investing (I), or financing (F) activities.

The balance sheet accounts related to earning income (operating items) should be marked with an O. These accounts are often called **operating assets and liabilities.** The accounts that should be marked with an O include the following:

• Most current assets (other than short-term investments, which relate to investing activities, and cash).

• Most current liabilities (other than amounts owed to investors and financial institutions, all of which relate to financing activities).

• Retained Earnings, because it increases by the amount of net income, which is the starting point for the operating section. (Retained Earnings also decreases by dividends declared and paid, which is a financing outflow noted by an F.)

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Exhibit 12.3 Comparative Balance Sheet

NATIONAL BEVERAGE CORP.
Consolidated Balance Sheets
(dollars in thousands)

Related Cash Flow Section Change in Cash		April 29 2023	April 30, 2022	*Change*
	Assets			
	Current assets:			
I	Cash and equivalents	$ 158,074	$ 48,050	+ 110,024
O	Short-term investments	5,248	5,015	+ 233
O	Accounts receivable	104,918	93,592	+ 11,326
O	Inventories	93,578	103,318	– 9,740
	Prepaid expenses	9,835	29,560	– 19,725
	Total current assets	371,653	279,535	
I†	Property, plant, and equipment—net	202,689	188,269	+ 14,420
	Total assets	$ 574,342	$ 467,804	
	Liabilities and Stockholders' Equity			
	Current liabilities:			
O	Accounts payable	$ 85,106	$ 95,299	– 10,193
O	Accrued expenses	59,215	50,020	+ 9,195
	Total current liabilities	144,321	145,319	
F	Long-term debt	37,720	59,244	– 21,504
O	Deferred income taxes—net	19,814	23,823	– 4,009
	Total liabilities	201,855	228,366	
	Stockholders' equity:			
F	Common stock	1,167	1,167	+ 0
F	Additional paid-in capital	16,160	15,172	+ 988
O and F	Retained earnings	355,160	223,099	+ 132,061
	Total stockholders' equity	372,487	239,438	
	Total liabilities and stockholders' equity	$ 574,342	$ 467,804	

†The Accumulated Depreciation account also is related to operations because it relates to depreciation.

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Exhibit 12.3 Current Income Statement

NATIONAL BEVERAGE CORP.
Consolidated Statement of Income*
For the Fiscal Year Ended April 29, 2023
(dollars in thousands)

Net sales	$ 1,172,932
Cost of sales	776,143
Gross profit	396,789
Operating expenses:	
Selling, general, and administrative expenses	202,573
Depreciation and amortization expense	7,532
Total operating expenses	210,105
Operating Income	186,684
Interest expense–net	242
Income before provision for income taxes	186,442
Provision for income taxes	44,278
Net income	$ 142,164

**Certain amounts have been adjusted for pedagogical purposes.*

Under the indirect method, the statement of cash flows begins with net income from the income statement.

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Learning Objective 12-2

12-2 Report and interpret cash flows from operating activities using the indirect method.

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Reporting and Interpreting Cash Flows from Operating Activities

The indirect method starts with net income and converts it to cash flows from operating activities.

This involves adjusting net income for the differences in the timing of accrual basis net income and cash flows.

Operating Activities
Net income
Adjustments to reconcile net income to cash flow from operating activities:
+ Depreciation and amortization expense
– Gain on sale of investing assets
+ Loss on sale of investing assets
+ Decreases in operating assets
+ Increases in operating liabilities
– Increases in operating assets
– Decreases in operating liabilities
Net Cash Flow from Operating Activities

Tip The indirect method operating activities section starts with net income and removes revenues and expenses that do not affect cash.

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Exhibit 12.4 (1 of 3)

Schedule for Net Cash Flow from Operating Activities, Indirect Method

CONVERSION OF NET INCOME TO NET CASH FLOW FROM OPERATING ACTIVITIES		
Items	**Amount**	**Explanation**
Net income, accrual basis	$ 142,164	From income statement.
Add (subtract) to convert to cash basis:		
Depreciation and amortization	7,532	Add back because depreciation and amortization expense does not affect cash.
Accounts receivable increase	(11,326)	Subtract because cash collected from customers is less than accrual basis revenues.
Inventory decrease	9,740	Add because purchases are less than cost of goods sold expense.
Prepaid expenses decrease	19,725	Add because cash prepayments for expenses are less than accrual basis expenses.
Accounts payable decrease	(10,193)	Subtract because cash payments to suppliers are more than amounts purchased on account (borrowed from suppliers).
Accrued expenses increase	9,195	Add because cash payments for expenses are less than accrual basis expenses.
Deferred income tax decrease	(4,009)	Subtract because cash payments for income taxes are more than income tax expense.
Net cash provided by operating activities	$ 162,828	Subtotal on the statement of cash flows.

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Adjust Net Income Depreciation and Amortization Expense

Recording depreciation and amortization expense does not affect the cash account (or any other operating asset or liability). It affects a noncurrent investing asset (Property, plant, and equipment, net). **Because depreciation and amortization expense is subtracted in computing net income but does not affect cash, we always add it back** to convert net income to cash flow from operating activities.

Depreciation Expense →	Add to net income.
Amortization Expense →	Add to net income.

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Adjust Net Income for Gains and Losses on Sale of Investing Assets

If property, plant, and equipment is sold at a gain or loss, the amount of cash received would be classified as an investing cash inflow.

Because all of the cash received is an investing cash flow, an adjustment also must be made in the operating activities section to avoid double counting the gain or loss.

Gains → Gains must be subtracted from net income to avoid double counting the gain.

Losses → Losses must be added to net income to avoid double counting the loss.

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Exhibit 12.4 (2 of 3)

Schedule for Net Cash Flow from Operating Activities, Indirect Method

CONVERSION OF NET INCOME TO NET CASH FLOW FROM OPERATING ACTIVITIES		
Items	**Amount**	**Explanation**
Net income, accrual basis	$ 142,164	From income statement.
Add (subtract) to convert to cash basis:		
Depreciation and amortization	7,532	Add back because depreciation and amortization expense does not affect cash.
Accounts receivable increase	(11,326)	Subtract because cash collected from customers is less than accrual basis revenues.
Inventory decrease	9,740	Add because purchases are less than cost of goods sold expense.
Prepaid expenses decrease	19,725	Add because cash prepayments for expenses are less than accrual basis expenses.
Accounts payable decrease	(10,193)	Subtract because cash payments to suppliers are more than amounts purchased on account (borrowed from suppliers).
Accrued expenses increase	9,195	Add because cash payments for expenses are
Deferred income t		me
		se.
Net cash provided by		

Step 1

Adjust net income for depreciation and amortization expense and gains and losses on sale of investing assets.

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Exhibit 12.4 (3 of 3)

Schedule for Net Cash Flow from Operating Activities, Indirect Method

CONVERSION OF NET INCOME TO NET CASH FLOW FROM OPERATING ACTIVITIES		
Items	**Amount**	**Explanation**
Net income, accrual basis	$ 142,164	From income statement.
Add (subtract) to convert to cash basis:		
Depreciation and amortization	7,532	Add back because depreciation and amortization expense does not affect cash.
Accounts receivable increase	(11,326)	Subtract because cash collected from customers is less than accrual basis revenues.
Inventory decrease	9,740	Add because purchases are less than cost of goods sold expense.
Prepaid expenses decrease	19,725	Add because cash prepayments for expenses are less than accrual basis expenses.
Accounts payable decrease	(10,193)	Subtract because cash payments to suppliers are more than amounts purchased on accou
Accrued expenses increase	9,195	Add beca less th
Deferred income tax decrease	(4,009)	Subtract taxes
Net cash provided by operating activities	$ 162,828	Subtotal

Step 2

Adjust net income for changes in current assets and current liabilities.

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Change in Accounts Receivable

Remember that the **income statement** reflects sales **revenue**, but the **cash flow statement** must reflect **cash** collections from customers.

When sales revenues are recorded, accounts receivable increases, and when cash is collected from customers, accounts receivable decreases.

	Accounts Receivable (A)			
Change $11,326	Beginning balance	93,592		
	Sales revenue (on account)	1,172,932	Collections from customers	1,161,606
	Ending balance	104,918		

- The balance sheet indicates an **increase** in accounts receivable of $11,326 for the period, which means that cash collected from customers is lower than revenue.

Ending balance	$104,918
– Beginning balance	93,592
Change	$ 11,326

- To convert to cash flows from operating activities, the amount of the **increase** (the amount of sales over and above collections) must be **subtracted** from net income. (A decrease is added.)

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Change in Inventory

The **income statement** reflects **merchandise sold** for the period, whereas **cash flow** from operating activities must reflect **cash** purchases.

Purchases of goods increase the balance in inventory, and recording merchandise sold decreases the balance in inventory.

Inventories (A)	
Beg. bal.	
Purchases	Costs of goods sold
End. bal.	

Inventories (A)			
Beg.	103,318		
		Decrease	9,740
End.	93,578		

- The balance sheet indicates that inventory **decreased** by $9,740, which means that the amount of purchases is less than the amount of merchandise sold.
- The **decrease** (the insufficient goods purchased) must be **added to** net income to convert to cash flow from operating activities (An increase is subtracted.)

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Change in Prepaid Expenses

The **income statement** reflects **expenses** of the period, but **cash flow** from operating activities must reflect the **cash** payments.

Cash prepayments increase the balance in prepaid expenses, and recording of expenses decreases the balance in prepaid expenses.

Prepaid Expenses (A)	
Beg. bal.	
Cash prepayments	Services used (expense)
End. bal.	

Prepaid Expenses (A)			
Beg.	29,560		
		Decrease	19,725
End.	9,835		

- The balance sheet indicates a $19,725 **decrease** in prepaid expenses, which means that new cash prepayments are less than the amount of expenses.
- The **decrease** (the extra prepayments) must be **added to** net income. (An increase is subtracted.)

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Change in Accounts Payable

Cash flow from operations must reflect cash purchases, but **not all purchases are for cash.**

Purchases on account increase accounts payable and cash paid to suppliers decreases accounts payable.

Accounts Payable (L)	
Beg. bal.	
Cash payments	Purchases on account
End. bal.	

Accounts Payable (L)			
		Beg.	95,299
Decrease	10,193		
		End.	85,106

- The balance sheet indicates accounts payable **decreased** by $10,193, which means that cash payments were more than purchases on account.
- This **decrease** (the extra purchases on account) must be **subtracted** from net income. (An increase is added.)

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Change in Accrued Expenses

The income statement reflects all accrued expenses, but the cash flow statement must reflect actual payments for those expenses.

Recording accrued expenses increases the balance in the liability accrued expenses and cash payments for the expenses decrease accrued expenses.

Accrued Expenses (L)	
	Beg. bal.
Cash payments	Accrued expenses
	End. bal.

Accrued Expenses (L)		
	Beg.	50,020
	Increase	9,195
	End.	59,215

- The balance sheet indicates accrued expenses **increased** by $9,195, which indicates that cash paid for the expenses is less than accrual basis expenses.
- The **increase** (the lower cash paid) must be **added** to net income. (A decrease is subtracted.)

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Change in Deferred Income Taxes - Net

The **income statement** reflects the total provision for income taxes (income tax **expense**), but the **cash flow statement** must reflect **actual payments** for income taxes during the period.

Tax payments that are postponed until future periods increase the deferred income taxes account and payments in the current period for prior period income taxes decrease the deferred income taxes account.

Deferred Income Taxes—Net (L)	
	Beg. bal
Pay prior years	Postpone current year
	End. bal

Deferred Income Taxes—Net (L)			
		Beg.	23,823
Decrease	4,009		
		End.	19,814

- The Deferred Income Taxes **decreased** by $4,009, which indicates that cash paid for income taxes is more than accrual basis income tax expense.
- The **decrease** (the additional taxes paid) must be **subtracted** from net income. (An increase is added.)

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Summary

We can summarize the typical additions and subtractions that are required to reconcile net income with cash flow from operating activities as follows:

	ADDITIONS AND SUBTRACTIONS TO RECONCILE NET INCOME TO CASH FLOW FROM OPERATING ACTIVITIES	
Item	When Item Increases	When Item Decreases
Depreciation and amortization	Add	NA
Gain on sale of long-term assets	Subtract	NA
Loss on sale of long-term assets	Add	NA
Accounts receivable	Subtract	Add
Inventory	Subtract	Add
Prepaid expenses	Subtract	Add
Accounts payable	Add	Subtract
Accrued expense liabilities	Add	Subtract
Deferred income tax liability	Add	Subtract

Tip When preparing the indirect method operating section, (1) add operating asset decreases and operating liability increases, and (2) subtract operating asset increases and operating liability decreases.

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Classification of Interest on the Cash Flow Statement

INTERNATIONAL PERSPECTIVE

U.S. GAAP and IFRS differ in the cash flow statement treatment of interest received and interest paid.

	Interest Received	Interest Paid
U.S. GAAP	Operating	Operating
IFRS	Operating or Investing	Operating or Financing

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Interpreting Cash Flows from Operating Activities

A common rule of thumb followed by financial and credit analysts is to avoid firms with rising net income but falling cash flow from operations.

In the long run, operations are the only source of cash!

Investors will not invest in a company if they do not believe that cash generated from operations will be available to pay them dividends or expand the company.

Creditors will not lend money if they do not believe that cash generated from operations will be available to pay back the loan.

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Learning Objective 12-3

12-3 Analyze and interpret the quality of income ratio.

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Quality of Income Ratio

KEY RATIO ANALYSIS

Many analysts compute the quality of income ratio as a general sign of the ability to generate cash through operations. **This ratio measures the portion of income that was generated in cash.** All other things equal, a higher quality of income ratio indicates greater ability to finance operating and other cash needs from operating cash inflows.

$$\text{Quality of Income Ratio} = \frac{\text{Cash Flow from Operating Activities}}{\text{Net Income}}$$

National Beverage Corp.'s ratio for the year 2023 was

$$\frac{\$162{,}828}{\$142{,}164} = 1.15\ (115\%)$$

COMPARISONS OVER TIME		
National Beverage		
2021	2022	2023
1.11	0.84	1.15

COMPARISONS WITH COMPETITORS	
Coca-Cola	PepsiCo
2023	2023
1.08	1.47

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Fraud and Cash Flow from Operations (1 of 2)

A QUESTION OF ETHICS

The cash flow statement often gives outsiders the first hint that financial statements may contain errors and irregularities.

Unethical managers sometimes attempt to reach earnings targets by manipulating accruals and deferrals of revenues and expenses to inflate income. Because these adjusting entries do not affect the cash account, they have no effect on the cash flow statement.

A growing difference between net income and cash flow from operations can be a sign of such manipulations.

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Fraud and Cash Flow from Operations (2 of 2)

A QUESTION OF ETHICS

Investors Chronicle reported on an accounting fraud at a commercial credit company, suggesting the following:

> . . . a look at **Versailles**'s cash flow statement—an invaluable tool in spotting creative accounting—should have triggered misgivings. In the company's last filed accounts . . . Versailles reported operating profits of . . . $25 million but a cash outflow from operating activities of $24 million . . . such figures should . . . have served as a warning. After all, what use is a company to anyone if it reports only accounting profits which are never translated into cash?

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Learning Objective 12-4

12-4 Report and interpret cash flows from investing activities.

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Reporting and Interpreting Cash Flows from Investing Activities

Preparing the Investing section of the cash flow statement requires an analysis of the accounts related to property, plant, and equipment; intangible assets; and investments in the securities of other companies.

Related Balance Sheet Account(s)	Investing Activity	Cash Flow Effect
Property, plant, and equipment and intangible assets (patents, etc.)	Purchase of property, plant, and equipment or intangible assets for cash	Outflow
	Sale of property, plant, and equipment or intangible assets for cash	Inflow
Short- or long-term investments (stocks and bonds of other companies)	Purchase of investment securities for cash	Outflow
	Sale (maturity) of investment securities for cash	Inflow

Remember that:

- Only purchases paid for with cash or cash equivalents are included.
- The amount of cash that is received from the sale of assets is included, regardless of whether the assets are sold at a gain or loss.

Tip Remember that the investing and financing activities are always presented using the "direct method." That is, they list the gross receipts and payments due to each activity.

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Exhibit 12.5 Schedule for Net Cash Flow from Investing Activities

Items	Cash Inflows (Outflows)	Explanation
Cash flows from investing activities:		
Purchases of property, plant, and equipment	$(21,979)	Payment in cash for equipment
Proceeds from disposal of property, plant, and equipment	27	Receipt of cash from sale of equipment
Purchase of short-term investments	(1,252)	Payment in cash for new investments
Proceeds from sale of short-term investments	1,019	Receipt of cash from sale of investments
Net cash used by investing activities	$(22,185)	

We must report individually the cash used to **purchase** equipment and the cash proceeds received from the **sale** of equipment.

Although short-term investments is a current asset, it is reported in the investing section on the statement of cash flows. The company **purchased** short-term investments for $1,252. They company also **sold** short-term investments for $1,019.

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Learning Objective 12-5

12-5 Analyze and interpret the capital acquisitions ratio.

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Capital Acquisitions Ratio

KEY RATIO ANALYSIS

This ratio reflects the portion of purchases of property, plant, and equipment financed from operating activities. A high ratio indicates less need for outside financing for current and future expansions.

$$\text{Capital Acquisitions Ratio} = \frac{\text{Cash Flow from Operating Activities}}{\text{Cash Paid for Property, Plant, and Equipment}}$$

National Beverage's ratio for 2023 was

$$\frac{\$162{,}828}{\$21{,}979} = 7.41$$

COMPARISONS OVER TIME		
National Beverage		
2021	2022	2023
7.66	4.59	7.41

COMPARISONS WITH COMPETITORS	
Coca-Cola	PepsiCo
2023	2023
6.26	2.44

A high ratio benefits the company because:

- it provides the company opportunities for strategic acquisitions
- it avoids the cost of additional debt
- it reduces the risk of bankruptcy that comes with additional leverage

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Free Cash Flow

FINANCIAL ANALYSIS

Managers and analysts often calculate free cash flow as a measure of a firm's ability to pursue long-term investment opportunities.

$$\text{Free Cash Flow} = \text{Cash Flow from Operating Activities} - \text{Dividends} - \text{Capital Expenditures}$$

- Any positive free cash flow is available for additional capital expenditures, investments in other companies, and mergers and acquisitions without the need for external financing or reductions in dividends to shareholders.
- Sometimes managers use free cash flow to pursue unprofitable investments just for the sake of growth or to obtain perquisites.

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Learning Objective 12-6

12-6 Report and interpret cash flows from financing activities.

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Reporting Cash Flows from Financing Activities

Financing activities are associated with generating capital from creditors and owners.

Related Balance Sheet Account(s)	Financing Activity	Cash Flow Effect
Short-term debt (notes payable)	Borrowing cash from banks or other financial institutions	Inflow
	Repayment of loan principal	Outflow
Long-term debt	Issuance of bonds for cash	Inflow
	Repayment of bond principal	Outflow
Common stock and additional paid-in capital	Issuance of stock for cash	Inflow
	Repurchase (retirement) of stock with cash	Outflow
Retained earnings	Payment of cash dividends	Outflow

Remember that:

- Cash repayments of principal are financing activities.
- Interest payments are operating activities.
- Dividend payments are cash flows from financing activities.
- If debt or stock is issued for other than cash, it is not included in this section.

Tip Interest payments are operating cash flows while dividend payments are financing cash flows.

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Exhibit 12.6

Items	Cash Inflows (Outflows)	Explanation
Cash flows from financing activities:		
Repayment of principal on long-term debt	(30,000)	Cash payments of principal on long-term debt
Proceeds from issuance of long-term debt	8,496	Cash proceeds from issuing long-term debt
Repurchase of stock	–	Cash payments to repurchase common stock
Proceeds from issuance of stock	988	Cash proceeds from issuing common stock
Payment of cash dividends	(10,103)	Cash payments of dividends to shareholders
Net cash used by financing activities	(30,619)	Subtotal on the statement of cash flows

Long-Term Debt (L)			
		Beginning	59,224
Retire (repay)	30,000	Issue (borrow)	8,496
		Ending	37,720

Retained Earnings (SE)			
		Beginning	223,099
Dividends	10,103	Net Income	142,164
		Ending	355,160

Common Stock (SE)			
		Beginning	1,167
Repurchase	–	Issue	–
		Ending	1,167

Additional Paid-in Capital (SE)			
		Beginning	15,172
Repurchase		Issue	988
		Ending	16,160

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Interpreting Cash Flows from Financing Activities

The long-term growth of a company is normally financed from three sources:

1. **Internally generated funds (cash from operating activities)**
2. **The issuance of stock**
3. **Money borrowed on a long-term basis**

The statement of cash flows shows how management has elected to fund its growth. This information is used by analysts who wish to evaluate the capital structure and growth potential of a business.

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Learning Objective 12-7

12-7 Understand the format of the cash flow statement and additional cash flow disclosures.

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Exhibit 12.7

Structure of the Statement of Cash Flows (Indirect Method)

Statement of Cash Flows (Indirect Method)
Operating Activities:
Net Income
+ Depreciation and amortization expense
– Gain on sale of long-term assets
+ Loss on sale of long-term assets
+ Decreases in operating assets
+ Increases in operating liabilities
– Increases in operating assets
– Decreases in operating liabilities
Net Cash Flow from Operating Activities
Investing Activities:
– Purchase of property, plant, and equipment or intangible assets
+ Sale of property, plant, and equipment or intangible assets
– Purchase of investment securities
+ Sale (maturity) of investment securities
Net Cash Flow from Investing Activities
Financing Activities:
+ Borrowing from bank or other financial institution
– Repayment of loan principal
+ Issuance of bonds for cash
– Repayment of bond principal
+ Issuance of stock
– Repurchase (retirement) of stock
– Payment of (cash) dividends
Net Cash Flow from Financing Activities
Net increase or decrease in cash and cash equivalents
Cash and cash equivalents at beginning of period
Cash and cash equivalents at end of period

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Supplemental Cash Flow Information

Two additional required cash flow disclosures are normally listed at the bottom of the statement or in the notes.

1. **Noncash investing and financing activities,** such as the purchase of a building with a mortgage given by the former owner.
2. **Cash paid for interest and cash paid for income taxes** (for companies that use the indirect method)

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Chapter Supplement A: Reporting Cash Flows from Operating Activities—Direct Method

	Sales revenue
+	Decrease in accounts receivable
–	Increase in accounts receivable
=	**Cash collected from customers**

	Interest/Dividend revenue
+	Decrease in interest/dividends receivable
–	Increase in interest/dividends receivable
=	**Collections of interest/dividends on investments**

	Cost of goods sold
+	Increase in inventory
–	Decrease in inventory
–	Increase in accounts payable
+	Decrease in accounts payable
=	**Cash payments to suppliers**

	Other expenses
+	Increase in prepaid expenses
–	Decrease in prepaid expenses
–	Increase in accrued expenses
+	Decrease in accrued expenses
=	**Cash paid for expenses**

	Income tax expense
+	Increase in prepaid income taxes
–	Decrease in prepaid income
–	Increase in income taxes payable
+	Decrease in income taxes
=	**Income taxes paid**

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Chapter Supplement B: Adjustment for Gains and Losses on Sale of Long-Term Assets—Indirect Method (1 of 2)

The Operating Activities section of the cash flow statement prepared using the indirect method may include an adjustment for gains and losses on the sale of long-term assets reported on the income statement.

If property, plant, and equipment with an original cost of $10,000 and accumulated depreciation of $4,000 is sold for $8,000 cash, the following entry is made.

$8,000 Investing cash inflow

$2,000 subtraction in the Operating section

	Debit	Credit
Cash (+A)	8,000	
Accumulated depreciation (−XA, +A)	4,000	
Property, plant, and equipment (−A)		10,000
Gain on disposal (+R, +SE) *($8,000 − $6,000)*		2,000

Assets		=	Liabilities	+	Stockholders' Equity	
Cash	+8,000				Gain on disposal (+R)	+2,000
Accumulated depreciation	+4,000					
Property, plant, and equipment	−10,000					

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Chapter Supplement B: Adjustment for Gains and Losses on Sale of Long-Term Assets—Indirect Method (2 of 2)

If property, plant, and equipment with an original cost of $10,000 and accumulated depreciation of $4,000 is sold for $5,000 cash, the following entry is made.

$5,000 Investing cash inflow

$1,000 addition in the Operating section

	Debit	Credit
Cash (+A)	5,000	
Accumulated depreciation (−XA, +A)	4,000	
Loss on disposal (+E, −SE) ($5,000 − $6,000)	1,000	
Property, plant, and equipment (−A)		10,000

Assets		=	Liabilities	+	Stockholders' Equity	
Cash	+5,000				Loss on disposal (+E)	−1,000
Accumulated depreciation	+4,000					
Property, plant, and equipment	−10,000					

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Chapter Supplement C: T-Account Approach (Indirect Method)

Based on the idea that changes in cash must equal the sum of the changes in all other balance sheet accounts, we can use T-accounts to analyze cash flows as follows:

1. Determine the change in each balance sheet account. From this year's ending balance, subtract this year's beginning balance (i.e., last year's ending balance).

2. Identify the cash flow category or categories to which each account relates.

3. Create schedules that summarize operating, investing, and financing cash flows.

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Exhibit 12.9 (1 of 2)

T-Account Approach to Preparing the Statement of Cash Flows (Indirect Method)

Panel A: Changes in Cash Account

Cash (A)			
Operating			
(1) Net Income	142,164	11,326	(3) Accounts Receivable
(2) Depreciation and Amortization	7,532	10,193	(6) Accounts Payable
(4) Inventory	9,740	4,009	(14) Deferred Income Taxes
(5) Prepaid Expenses	19,725		
(7) Accrued Expenses	9,195		
Net cash flow provided by operating activities	162,828		
Investing			
(9) Disposals of Property, Plant, and Equipment	27	21,979	(8) Purchases of Property, Plant, and Equipment
(11) Sales of Short-Term Investments	1,019	1,252	(10) Purchases of Short-Term Investments
		22,185	Net cash used in investing activities
Financing			
(13) Issuance of Long-Term Debt	8,496	30,000	(12) Retire Long-Term Debt
(15) Issuance of Stock	988	10,103	(16) Payment of Dividends
		30,619	Net cash used in financing activities
Net decrease in cash and cash equivalents		(110,024)	

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Exhibit 12.9 (2 of 2)

T-Account Approach to Preparing the Statement of Cash Flows (Indirect Method)

Panel B: Changes in Noncash Accounts

Accounts Receivable (A)			
Beg. bal.	93,592		
(3) Increase	11,326		
End. bal.	104,918		

Inventories (A)			
Beg. bal.	103,318		
		(4) Decrease	9,740
End. bal.	93,578		

Prepaid Expenses (A)			
Beg. bal.	29,560		
		(5) Decrease	19,725
End. bal.	9,835		

Accounts Payable (L)			
		Beg. bal.	95,299
(6) Decrease	10,193		
		End. bal.	85,106

Accrued Expenses (L)			
		Beg. bal.	50,020
		(7) Increase	9,195
		End. bal.	59,215

Property, Plant and Equipment, Net (A)			
Beg. bal.	188,269	(2) Depreciation	7,532
(8) Purchases	21,979	(9) Disposals	27
End. bal.	202,689		

Short-Term Investments (A)			
Beg. bal.	5,015		
(10) Purchases	1,252	(11) Disposals	1,019
End. bal.	5,248		

Long-Term Debt (L)			
		Beg. bal.	59,224
(12) Retire (repayments)	30,000	(13) Issue (borrowings)	8,496
		End. bal.	37,720

Deferred Income Taxes, Net (L)			
		Beg. bal.	23,823
(14) Decrease	4,009		
		End. bal.	19,814

Common Stock (SE)			
		Beg. bal.	1,167
		End. bal.	1,167

Additional Paid-in Capital (SE)			
		Beg. bal.	15,172
		(15) Stock issued	988
		End. bal.	16,160

Retained Earnings (SE)			
		Beg. bal.	223,099
(16) Dividends	10,103	(1) Net income	142,164
		End. bal.	355,160

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